

Chapter 2 Review • KEY**2.1 Conditional Statements**

1. Rewrite each statement in if-then form.

a) All right angles are congruent.

→ If two angles are right angles, then they are congruent.

b) I go to practice whenever school is in session.

→ If school is in session, then I go to practice.

2. Identify the hypothesis and the conclusion: If a polygon has five sides, then it is a pentagon.

Hypothesis: _____ Conclusion: _____

→ H: a polygon has five sides; C: it is a pentagon

3. Write the converse, inverse, and contrapositive of "If a number is even, then it is divisible by 2." Then tell whether each is true or false.

Converse:

→ If a number is divisible by 2, then it is even. TRUE

Inverse:

→ If a number is not even, then it is not divisible by 2. TRUE

Contrapositive:

→ If a number is not divisible by 2, then it is not even. TRUE

Since the conditional and its converse are both true, write the biconditional:

→ A number is even if and only if it is divisible by 2.

4. Write the negation.

a) The angle is obtuse.

→ The angle is not obtuse.

b) The lines are not parallel.

→ The lines are parallel.

5. Can "If a number is divisible by 5, then it ends in 0" be written as a biconditional? Explain.

→ No. The conditional itself is false (15 is divisible by 5 and ends in 5), so there is no true biconditional.

6. Use the diagram. Decide whether each statement is true or false. Explain.

a) $\angle AMC$ and $\angle CMB$ form a linear pair.

TRUE / FALSE because

→ TRUE — ray MA and ray MB are opposite rays.

b) $\angle AMC \cong \angle BMD$

TRUE / FALSE because

→ TRUE — they are vertical angles.

c) Ray MA and ray MC are opposite rays.

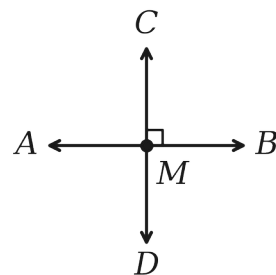
TRUE / FALSE because

→ FALSE — they do not form a straight line.

d) $\angle AMD$ is acute.

TRUE / FALSE because

→ FALSE — the right-angle mark shows the lines are $\perp$, so $\angle AMD$ is a right angle (90°).



2.2 Inductive & Deductive Reasoning**7.** Find the next two items in each pattern.a) 3, 7, 11, 15, ... ,
 4, ... ,

b) 1, 2, 4, 8, ... ,

c) -5, -2, 1,

→ a) 19, 23 (add 4) b) 16, 32 (multiply by 2) c) 7, 10 (add 3)

8. Find a counterexample for each conjecture.

a) The square of a number is always greater than the number.

→ $1^2 = 1$, not greater ($\frac{1}{2}$ or 0 also work)

b) If two angles are supplementary, then they form a linear pair.

→ Two separate 90° angles drawn apart are supplementary but not adjacent, so not a linear pair.

c) Every quadrilateral has four right angles.

→ A parallelogram that is not a rectangle (or any rhombus that is not a square).

9. Inductive or deductive? Circle one and explain.

a) Every time Coach blows the whistle twice, practice ends. He just blew it twice, so practice is over.

INDUCTIVE / DEDUCTIVE because

→ INDUCTIVE — a pattern of past cases, not a rule.

b) All squares are rectangles. ABCD is a square. So ABCD is a rectangle.

INDUCTIVE / DEDUCTIVE because

→ DEDUCTIVE — a general rule applied by logic (Law of Detachment).

10. Is each conjecture valid? If valid, name the law. If not, explain.

a) If a bird is a penguin, then it cannot fly. Pete is a penguin. Conjecture: Pete cannot fly.

→ VALID — Law of Detachment.

b) If x is a multiple of 8, then x is even. The number 14 is even. Conjecture: 14 is a multiple of 8.→ NOT VALID — the conclusion being true does not make the hypothesis true ($14 = 8 \cdot 1.75$).

c) If a boy is in Room 406, then he is in Honors Geometry. If a boy is in Honors Geometry, then he writes proofs. Conjecture: If a boy is in Room 406, then he writes proofs.

→ VALID — Law of Syllogism ($p \rightarrow q$ and $q \rightarrow r$ give $p \rightarrow r$).**2.3 Postulates & Diagrams****11.** Name the postulate.

a) Points R and S lie in plane V, so line RS lies in plane V.

→ Plane-Line Postulate

b) Lines m and n intersect at exactly one point.

→ Line Intersection Postulate

c) Through points A, B, and C (noncollinear) there is exactly one plane.

→ Three Point Postulate

d) Plane W contains at least three noncollinear points.

→ Plane-Point Postulate

12. Use the diagram. Is each statement valid? Circle and explain.

a) Points A, B, and C are collinear.

VALID / NOT VALID

→ VALID — all three are on the drawn line.

b) Line $EB \perp$ plane P.

VALID / NOT VALID

→ VALID — the right-angle mark at B.

c) Line $BD \perp$ line AC.

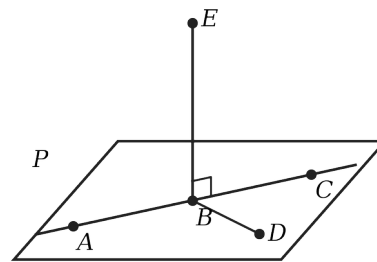
VALID / NOT VALID

→ NOT VALID — no mark between BD and AC.

d) Points B, D, and E are coplanar.

VALID / NOT VALID

→ VALID — any three points are coplanar (Three Point Postulate); they just do not all lie in plane P.



13. True or false? Explain. Two planes can intersect in exactly one point.

→ FALSE — if two planes intersect, their intersection is a line (Plane Intersection Postulate).

2.4 Algebraic Reasoning

14. Name the property that justifies each statement.

Statement	Property
$m\angle A = m\angle A$	Reflexive Property of Equality
If segment $GH \cong$ segment JK , then segment $JK \cong$ segment GH .	Symmetric Property of Congruence
If $AB = CD$ and $CD = EF$, then $AB = EF$.	Transitive Property of Equality
If $x = 4$, then $3x + 1 = 3(4) + 1$.	Substitution Property of Equality
$-7(x - 2) = -7x + 14$	Distributive Property
If $3x = 12$, then $x = 4$.	Division Property of Equality

15. Solve. Write every step and justify each one.

a) $5x - 8 = 2x + 7$

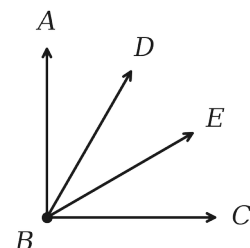
b) $-2(3y - 9) + 4y = 4$

Statements	Statements
1. $5x - 8 = 2x + 7$	1. $-2(3y - 9) + 4y = 4$
2. $3x - 8 = 7$	2. $-6y + 18 + 4y = 4$
3. $3x = 15$	3. $-2y + 18 = 4$
4. $x = 5$	4. $-2y = -14$
	5. $y = 7$

- 16.** Angle Addition proof. Reasons for lines 2 and 4 are done — supply the rest.

Given: $m\angle ABE = m\angle DBC$.

Prove: $m\angle ABD = m\angle EBC$.



Statements	Reasons
1. $m\angle ABE = m\angle DBC$	1. Given
2. $m\angle ABE = m\angle ABD + m\angle DBE$; $m\angle DBC = m\angle DBE + m\angle EBC$	2. Angle Addition Postulate
3. $m\angle ABD + m\angle DBE = m\angle DBE + m\angle EBC$	3. Substitution Property of Equality
4. $m\angle DBE = m\angle DBE$	4. Reflexive Property of Equality
5. $m\angle ABD = m\angle EBC$	5. Subtraction Property of Equality

For you: Line 4 is the Reflexive line — the shared piece $m\angle DBE$ named before it is subtracted. Same convention as 2.6.

2.5 Proving Statements about Segments and Angles

- 17.** Complete each statement.

a) Reflexive Property of Congruence: segment $PQ \cong$ _____ b) Symmetric: If $\angle A \cong \angle B$, then $\angle B \cong$ _____

c) Transitive: If segment $AB \cong$ segment CD and segment $CD \cong$ segment EF , then _____

→ a) segment PQ b) $\angle A$ c) segment $AB \cong$ segment EF

- 18.** M is the midpoint of segment JK . $JM = 9x - 4$ and $MK = 5x + 20$. Find x , JM , and JK .

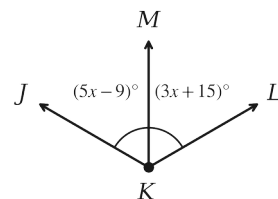
$x =$ _____ $JM =$ _____ $JK =$ _____

→ $9x - 4 = 5x + 20 \rightarrow 4x = 24 \rightarrow x = 6$; $JM = MK = 50$; $JK = 100$

- 19.** Complete the proof. Statements are done — supply each reason.

Given: Ray KM bisects $\angle JKL$; $m\angle JKM = (5x - 9)^\circ$, $m\angle MKL = (3x + 15)^\circ$.

Prove: $x = 12$



Statements	Reasons
1. Ray KM bisects $\angle JKL$; $m\angle JKM = (5x - 9)^\circ$, $m\angle MKL = (3x + 15)^\circ$	1. Given
2. $\angle JKM \cong \angle MKL$	2. Definition of angle bisector
3. $m\angle JKM = m\angle MKL$	3. Definition of congruent angles
4. $5x - 9 = 3x + 15$	4. Substitution Property of Equality
5. $2x - 9 = 15$	5. Subtraction Property of Equality
6. $2x = 24$	6. Addition Property of Equality
7. $x = 12$	7. Division Property of Equality

20. Overlapping segments. Statements are done — supply each reason.

Given: $AB = CD$. Prove: $AC = BD$.



Statements	Reasons
1. $AB = CD$	1. Given
2. $BC = BC$	2. Reflexive Property of Equality
3. $AB + BC = CD + BC$	3. Addition Property of Equality
4. $AC = AB + BC$; $BD = BC + CD$	4. Segment Addition Postulate
5. $AC = BD$	5. Substitution Property of Equality

2.6 Proving Geometric Relationships

21. Two lines intersect so that one of the angles measures 48° . Find the other three angle measures, and name the reason for each.

→ 132° (linear pair → Linear Pair Postulate), 48° (vertical → Vertical Angles Theorem), 132° (linear pair)

22. Find each value.

a) Vertical angles measure $(10v - 17)^\circ$ and 73° . Find v .

$v =$ _____

→ $10v - 17 = 73 \rightarrow 10v = 90 \rightarrow v = 9$

b) A linear pair measures $(7t + 12)^\circ$ and $(3t - 2)^\circ$. Find both measures.

_____ $^\circ$ and _____ $^\circ$

→ $10t + 10 = 180 \rightarrow t = 17$; 131° and 49°

23. What can you conclude? Name the theorem.

a) $\angle 1$ and $\angle 2$ are both complementary to $\angle 3$.

→ $\angle 1 \cong \angle 2$ — Congruent Complements Theorem

b) $\angle 4$ and $\angle 5$ are right angles.

→ $\angle 4 \cong \angle 5$ — Right Angles Congruence Theorem

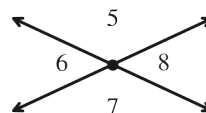
c) $\angle 6$ and $\angle 7$ form a linear pair.

→ $\angle 6$ and $\angle 7$ are supplementary — Linear Pair Postulate

24. Prove the Vertical Angles Theorem. Statements are done — supply each reason.

Given: $\angle 5$ and $\angle 7$ are vertical angles.

Prove: $\angle 5 \cong \angle 7$.



Statements	Reasons
1. $\angle 5$ and $\angle 7$ are vertical angles	1. Given
2. $\angle 5$ and $\angle 6$ form a linear pair; $\angle 6$ and $\angle 7$ form a linear pair	2. Definition of linear pair (from the diagram)
3. $\angle 5$ and $\angle 6$ are supplementary; $\angle 6$ and $\angle 7$ are supplementary	3. Linear Pair Postulate
4. $\angle 5 \cong \angle 7$	4. Congruent Supplements Theorem

25. The hard one. Congruent Complements when the shared angle is not the SAME angle — it is a congruent one. Lines 2 and 4 are done.

Given: $\angle A$ and $\angle B$ are complementary; $\angle C$ and $\angle D$ are complementary; $\angle B \cong \angle D$.

Prove: $\angle A \cong \angle C$



Statements	Reasons
1. $\angle A$ and $\angle B$ are complementary; $\angle C$ and $\angle D$ are complementary; $\angle B \cong \angle D$	1. Given
2. $m\angle A + m\angle B = 90^\circ$; $m\angle C + m\angle D = 90^\circ$	2. Definition of complementary angles
3. $m\angle A + m\angle B = m\angle C + m\angle D$	3. Substitution Property of Equality
4. $m\angle B = m\angle D$	4. Definition of congruent angles
5. $m\angle A + m\angle B = m\angle C + m\angle B$	5. Substitution Property of Equality
6. $m\angle B = m\angle B$	6. Reflexive Property of Equality
7. $m\angle A = m\angle C$	7. Subtraction Property of Equality
8. $\angle A \cong \angle C$	8. Definition of congruent angles

For you: Line 5 is the new move: replace $m\angle D$ with $m\angle B$ so both sides share the same piece — THEN the Reflexive line and the subtraction. Without line 5 the subtraction has nothing to cancel.

REASON BANK — every reason on the test comes from this list

Given · Definition of midpoint · Definition of a segment bisector · Definition of angle bisector · Definition of congruent angles · Definition of supplementary angles · Definition of complementary angles · Definition of right angle

Segment Addition Postulate · Angle Addition Postulate · Linear Pair Postulate · Vertical Angles Theorem · Congruent Supplements Theorem · Congruent Complements Theorem · Right Angles Congruence Theorem

Distributive Property · Simplify (combine like terms) · Addition / Subtraction / Multiplication / Division Property of Equality · Substitution Property of Equality

Reflexive / Symmetric / Transitive Property of Equality · Reflexive / Symmetric / Transitive Property of Congruence